

Assignment 4 Report

1. Objective

The purpose of this assignment was to extend a baseline Distance Vector Routing (DVR) implementation by introducing non-uniform link costs, implementing the Poisoned Reverse technique to prevent routing loops, and analysing the network's behaviour and convergence times during a simulated link failure. To thoroughly evaluate the routing logic, the implementation included a runtime toggle to observe network reconvergence both with and without Poisoned Reverse active.

2. Implementation of Weighted Routing

The baseline algorithm was successfully modified to parse and calculate routing paths based on variable link costs rather than a simple hop count.

Observations:

- Upon initialisation, the network accurately calculated the shortest paths based on the provided weights (A-B cost: 5, B-C cost: 10).
- Node A correctly determined that its shortest path to Node C was through Node B, resulting in a total accumulated cost of 15.
- The algorithm required **3 rounds** to reach initial saturation and converge.

```
Printing the routing tables after the convergence
A:
10.0.0.1 | 10.0.0.1 | 10.0.0.1 | 0
10.0.0.21 | 10.0.0.21 | 10.0.0.1 | 5
10.0.1.23 | 10.0.0.21 | 10.0.0.1 | 5
10.0.1.3 | 10.0.0.21 | 10.0.0.1 | 15
B:
10.0.0.21 | 10.0.0.21 | 10.0.0.21 | 0
10.0.1.23 | 10.0.1.23 | 10.0.1.23 | 0
10.0.0.1 | 10.0.0.1 | 10.0.0.21 | 5
10.0.1.3 | 10.0.1.3 | 10.0.1.23 | 10
C:
10.0.1.3 | 10.0.1.3 | 10.0.1.3 | 0
10.0.0.21 | 10.0.1.23 | 10.0.1.3 | 10
10.0.1.23 | 10.0.1.23 | 10.0.1.3 | 10
10.0.0.1 | 10.0.1.23 | 10.0.1.3 | 15
```

3. Simulation of Link Failure (With Poisoned Reverse)

To test the network's resilience, the direct link between Node A (10.0.0.21) and Node B (10.0.0.1) was manually severed by setting its link cost to 999 (representing infinity).

Observations:

- **Immediate Isolation:** Upon failure, Node A and Node B instantly updated their direct interface costs to 999.
- **Preventing the Loop:** Because Poisoned Reverse was active, Node B dynamically constructed neighbour-specific routing updates. When B sent its update to C, it advertised its cost to A as infinity.
- **Trusting the Next-Hop:** Node C, which previously relied on Node B to reach Node A, received this "poisoned" update. Instead of attempting to advertise its older, cheaper path back to B, Node C trusted its next-hop and correctly updated its own cost to Node A to 999.
- **Convergence Time:** The network cleanly handled the failure and fully reconverged in exactly 2 rounds.

```
--- SIMULATING LINK FAILURE (A to B) ---
--> Algorithm converged in 2 rounds.
Printing the routing tables after the convergence
A:
10.0.0.1 | 10.0.0.1 | 10.0.0.1 | 0
10.0.0.21 | 10.0.0.21 | 10.0.0.1 | 999
10.0.1.23 | 10.0.0.21 | 10.0.0.1 | 999
10.0.1.3 | 10.0.0.21 | 10.0.0.1 | 1009
B:
10.0.0.21 | 10.0.0.21 | 10.0.0.21 | 0
10.0.1.23 | 10.0.1.23 | 10.0.1.23 | 0
10.0.1.3 | 10.0.1.3 | 10.0.1.23 | 10
10.0.0.1 | 10.0.0.1 | 10.0.0.21 | 999
C:
10.0.1.3 | 10.0.1.3 | 10.0.1.3 | 0
10.0.0.21 | 10.0.1.23 | 10.0.1.3 | 10
10.0.1.23 | 10.0.1.23 | 10.0.1.3 | 10
10.0.0.1 | 10.0.1.23 | 10.0.1.3 | 999
```

4. Simulation of Link Failure (Without Poisoned Reverse)

To demonstrate the necessity of the protocol, the simulation was rerun with Poisoned Reverse disabled.

Observations:

- **The "Count-to-Infinity" Problem:** Without Poisoned Reverse, Node C was unaware that its path to A (which relied on B) was dead. Node C falsely advertised to Node B that it still had a valid path to A.
- **Routing Loop:** Node B, seeing its direct link to A was dead, accepted Node C's invalid path. Node B routed through C, and Node C routed through B.
- **Convergence Time:** The two nodes entered an infinite loop, continuously inflating the route cost in each round. The algorithm took **52 rounds** to finally converge, which happened only because the path cost eventually reached the artificial infinity ceiling of 999, forcing the nodes to drop the route.

```
--- SIMULATING LINK FAILURE (A to B) ---
--> Algorithm converged in 52 rounds.
Printing the routing tables after the convergence
A:
10.0.0.1 | 10.0.0.1 | 10.0.0.1 | 0
10.0.0.21 | 10.0.0.21 | 10.0.0.1 | 999
10.0.1.23 | 10.0.0.21 | 10.0.0.1 | 999
10.0.1.3 | 10.0.0.21 | 10.0.0.1 | 1009
B:
10.0.0.21 | 10.0.0.21 | 10.0.0.21 | 0
10.0.1.23 | 10.0.1.23 | 10.0.1.23 | 0
10.0.1.3 | 10.0.1.3 | 10.0.1.23 | 10
10.0.0.1 | 10.0.0.1 | 10.0.0.21 | 999
C:
10.0.1.3 | 10.0.1.3 | 10.0.1.3 | 0
10.0.0.21 | 10.0.1.23 | 10.0.1.3 | 10
10.0.1.23 | 10.0.1.23 | 10.0.1.3 | 10
10.0.0.1 | 10.0.1.23 | 10.0.1.3 | 999
```

5. Conclusion

The output confirms that the modified distance vector routing protocol functions correctly. The transition from hop-count to weighted links was successful. Furthermore, the comparative analysis of the link failure clearly demonstrates the critical importance of the Poisoned Reverse technique. By proactively advertising infinite costs to next-hop neighbours, the protocol successfully reduced convergence time from 52 rounds to just 2 rounds, effectively eliminating the count-to-infinity vulnerability.